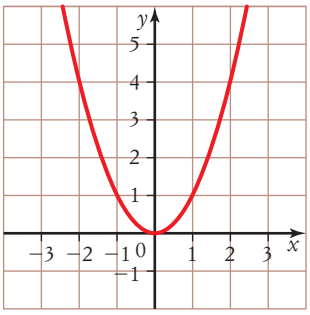
WORKSHEET

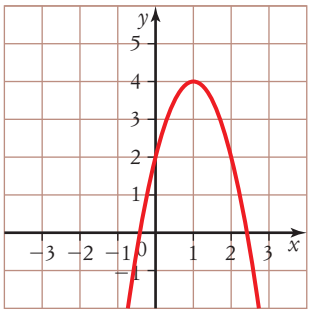
Concavity

1 State the values of x, if any, for which the following functions have $f''(x) < 0$, $f''(x) = 0$ and $f''(x) > 0$.

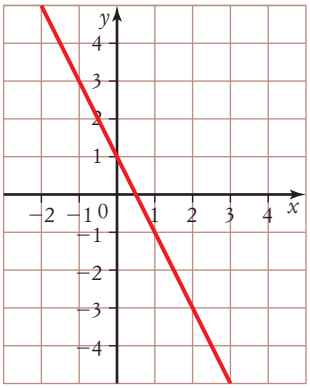
a $f(x) = x^2$



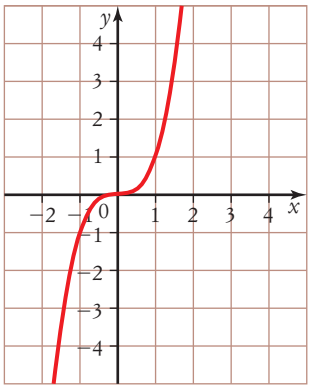
b $f(x) = -2x^2 + 4x + 2$



c $f(x) = -2x + 1$



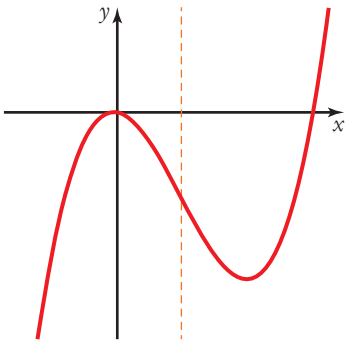
d $f(x) = x^3$



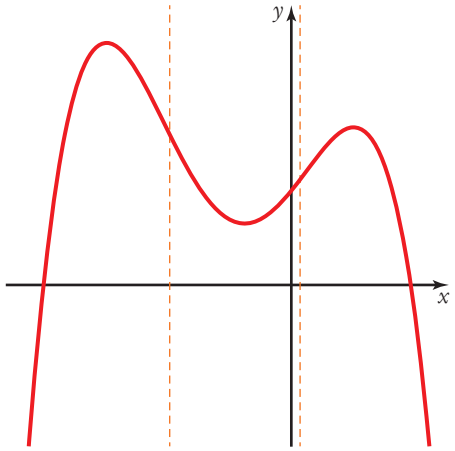
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2 In the following graphs the solutions of $f''(x) = 0$ are marked with a dashed line. Determine the sign of $f''(x)$ to the left and right of the dashed lines.

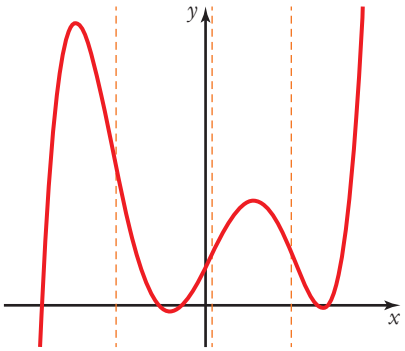
a



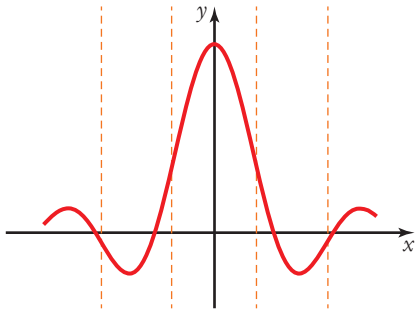
b



c



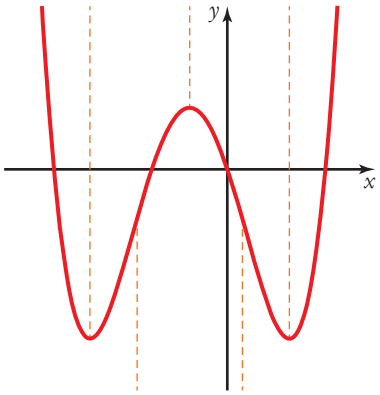
d



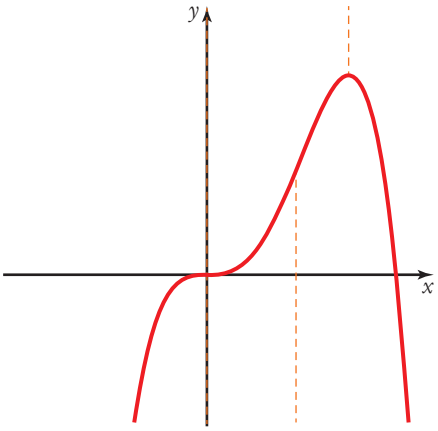
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3 Dashed lines in the graphs below indicate critical points. On either side of the dashed lines write the sign of the first derivative above the graph, and the sign of the second derivative below the graph.

a



b



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Answers

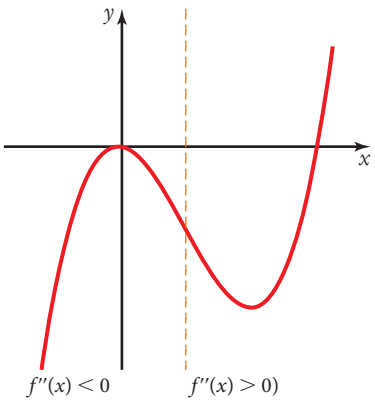
1 **a** $f''(x) > 0$ for all x

b $f''(x) < 0$ for all x

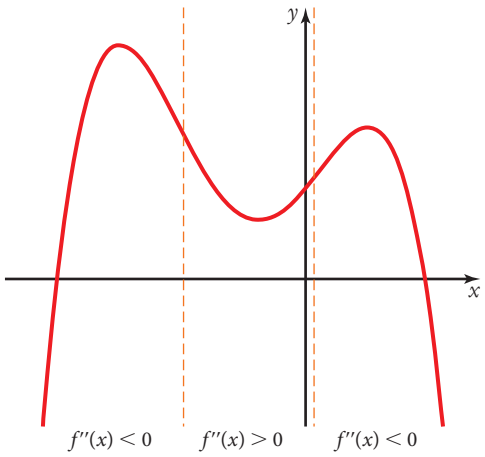
c $f''(x) = 0$ for all x

d $f''(x) < 0$ for $x < 0$ $\quad f''(x) = 0$ for $x = 0$ $\quad f''(x) > 0$ for $x > 0$

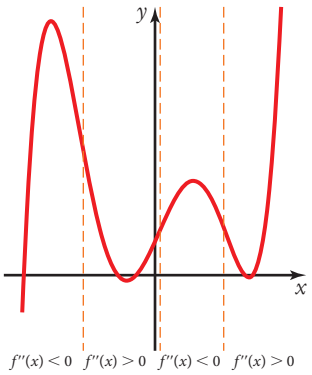
2 **a**



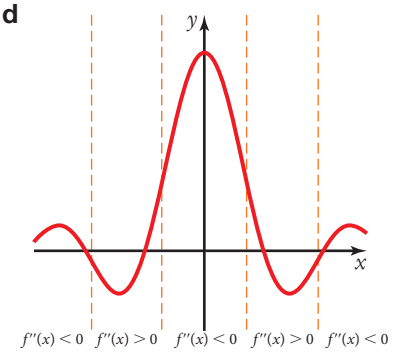
b



c

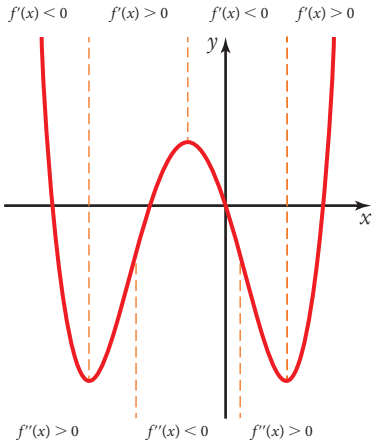


d

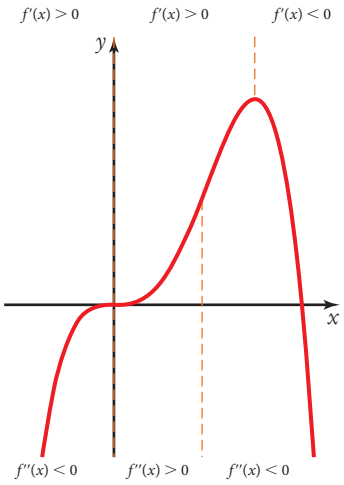


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3 a



b



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